

Tokens and Context

Project: Verified Study Buddy

1. Tokens and Context

Teach the learning cycle

Primary target: Account for fixed text, evidence and reserved output under a toy budget

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

A Distribution of Possibilities

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a next-token table is a valid probability distribution

Reference example, not the only acceptable wording: Teacher explanation: Entries are nonnegative and sum to 1. Not ready has mass $0.3+0.2 = 0.5$, including busy and unknown.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A toy system allows 12 tokens total for input and reserved output. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves evidence.

Concept to explain

Models operate on tokens within a limited context window.

Term · Meaning

Term: token

Meaning: A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.

Term: tokenization

Meaning: Converting text into the token units of a particular tokenizer.

Term: context window

Meaning: The bounded amount of token information a model can process together in a particular use.

Term: truncation

Meaning: Removing part of information, often because it exceeds a length or context limit.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Use two 45-minute sessions. All model outputs and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed. Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

No. Evaluate retrieval and claim support directly using sources and task-specific criteria.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

Apply the connected idea: Attention as weighted information selection

Use two 45-minute sessions. All model outputs and probability tables are constructed classroom examples. No hosted AI service, external API or student account is needed. Distinguish a model objective from factual support and system permissions. Collect independent work before discussing its solutions.

Readiness answer

Summation removes row order, so different ordered vectors may produce the same aggregate.

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Prepare the independent task and fresh retry without displaying solutions

End goal for the pictorial case

By the end, I can budget context and preserve evidence needed to interpret a requested fact.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. A sentence becomes token pieces, not a stored human thought.
2. The model uses visible tokens to predict the next token.
3. Remove the definition from context: the same question becomes ambiguous.
4. More context helps only when it is relevant, accurate, and within the window.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A toy window holds eight tokens; prompt uses six and quotation uses three.

Does it fit? What should be preserved?

Exact explanation to model

Nine exceeds eight. Shorten or select material while preserving essential instructions and evidence. Real systems also have output-budget rules.

Explain the added worked example

For this lesson only, a fictional system allows 100 total token units for input plus reserved output. Its fixed instructions use 20 units, question 10 and output reserve 20. That leaves 50 units for evidence. These are supplied counts, not word counts or a claim about any deployed model. Other systems can use different input/output accounting, so read their exact contract.

The question is: “Which room is approved for the robotics meeting, and what is its capacity?” Evidence A, a current approval notice, costs 30 units and names room B. Evidence B, the current room register, costs 20 and gives room B capacity 12. Evidence C, an old approval notice, costs 25 and names room A.

A+B costs 50 and fits exactly. Together they support both requested claims. A+C costs 55 and exceeds the evidence budget, while C+B fits at 45 but supplies an outdated room decision. A cheap set is not necessarily an adequate set. Preserve document IDs, versions and dates alongside content.

Key terms

Term · Meaning

Term: Context budget

Meaning: The permitted input and output allocation under a stated model or classroom contract

Term: Evidence selection

Meaning: Choosing source material relevant to the requested claim

Term: Reserved output

Meaning: Capacity deliberately kept available for the response

Term: Truncation

Meaning: Removing part of a sequence to meet a limit

Apply the connected idea: Attention as weighted information selection

Attention computes a weighted combination of value vectors, with weights derived from query-key compatibility. Queries, keys and values are distinct roles even when they come from the same sequence. A weight is a numerical contribution in this computation, not proof of the source's truth or permission. This week supplies weights directly; Week 7 calculates them from scores.

Use three values $V1=[2,0]$, $V2=[0,4]$, $V3=[4,2]$ with weights 0.5,0.25,0.25. Values have equal width and weights are nonnegative with sum 1. Compute each coordinate separately. No token is literally copied by this combination.

Adding the weighted vectors gives $[2,1.5]$. The same weights applied to different values can give a different result; the same values with different weights can also change it. For example weights 0,1,0 yield $V2=[0,4]$. The result is a representation to be used by later operations, not automatically a generated word.

Guided: $[6,0]/3 + 2*[0,3]/3 = [2,2]$. Masking B leaves $[6,0]$. A masked source is excluded from the stated information path. Exit: coefficient size does not verify a document or grant instruction authority.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A source may contain an exception after a general statement. Cutting off the last paragraph can reverse its practical meaning. Selection should preserve the passages needed to interpret the requested fact, including scope, version and exceptions. If complete relevant evidence cannot fit, retrieve a smaller authoritative passage or request clarification rather than silently remove a qualifying condition.

Including a document in context does not prove the generated answer uses it correctly. After generation, check each claim against its source. A source's instruction-like content also does not acquire authority just

because it fits in the evidence budget.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Tokens and Context

A fictional assistant has a small context budget for answering an event-capacity question.

The facts we will use

Total budget 100 toy tokens; instructions 20, question 10, output reserve 25. Evidence capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Context budget allocations

	Toy tokens
Instructions	20
Question	10
Output reserve	25
Evidence available	45

Read a cell using both its row and column labels.

Tokens and Context

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute budget

$100 - 20 - 10 - 25 = 45$ evidence tokens.



2. Select relevant pair

A+B costs 38 and preserves the capacity exception.



3. Reject incomplete shortcut

A+C costs 30 but leaves out the exception that changes the answer.



4. Answer with scope

For this event, the declared cap is 12, with both A and B retained.

CHANGE ONE THING

Increase B's cost to 30 with all other costs fixed.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Compute budget

What happens and why: $100 - 20 - 10 - 25 = 45$ evidence tokens.

Step: 2. Select relevant pair

What happens and why: A+B costs 38 and preserves the capacity exception.

Step: 3. Reject incomplete shortcut

What happens and why: A+C costs 30 but leaves out the exception that changes the answer.

Step: 4. Answer with scope

What happens and why: For this event, the declared cap is 12, with both A and B retained.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A context window limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement. Keep the relevant source relationship visible. Toy token costs are supplied here; real token counts depend on the model's tokenizer.

What this example does not establish

Fitting in context does not guarantee attention, correct interpretation or accurate generation.

Change one thing in the pictured case

Change: Increase B's cost to 30 with all other costs fixed.

Prediction: A+B costs 48 and no longer fits the 45-token evidence capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-evidence response.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A toy window holds eight tokens; prompt uses six and quotation uses three.

Write the question to explain: Does it fit? What should be preserved?

Reveal after predictions: Nine exceeds eight. Shorten or select material while preserving essential

instructions and evidence. Real systems also have output-budget rules.

Keep for an independent attempt: A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: $80-15-10-15 = 40$; $18+22$ fits. Claim support must still be checked. Exit: the register lists room B capacity 12, but current meeting approval is unavailable; request that approval before naming it as approved.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and distinguish a numerical claim from its interpretation.

Stage: 7 to 18 minutes

Teacher action: Read section 1 and complete W1; ask for a reason alongside each answer.

Stage: 18 to 32 minutes

Teacher action: Trace the worked case in section 2 and complete W2.

Stage: 32 to 45 minutes

Teacher action: Discuss the diagram and complete W3 in pairs; record support.

Stage: Session 2 start to 8 minutes

Teacher action: Complete the guided response before sharing the answer.

Stage: 8 to 20 minutes

Teacher action: Read the remaining sections and trace another constructed case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, requiring calculations and a bounded conclusion.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit response. Provide feedback and a fresh retry.

Teach the connected idea

Stage · Teacher action

Stage: Session 1 start to 7 minutes

Teacher action: Retrieve the prerequisite and distinguish a numerical claim from its interpretation.

Stage: 7 to 18 minutes

Teacher action: Read section 1 and complete W1; ask for a reason alongside each answer.

Stage: 18 to 32 minutes

Teacher action: Trace the worked case in section 2 and complete W2.

Stage: 32 to 45 minutes

Teacher action: Discuss the diagram and complete W3 in pairs; record support.

Stage: Session 2 start to 8 minutes

Teacher action: Complete the guided response before sharing the answer.

Stage: 8 to 20 minutes

Teacher action: Read the remaining sections and trace another constructed case.

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently, requiring calculations and a bounded conclusion.

Stage: 35 to 45 minutes

Teacher action: Discuss W5 and the exit response. Provide feedback and a fresh retry.

Reduce help deliberately

Completion task: Compare $A+B$, $A+C$, $C+B$ for fit and support of current approval and capacity.

Teacher check: Teacher explanation: $A+B$ costs 50 and supports both. $A+C$ costs 55 and fails budget; $C+B$ costs 45 but uses old approval and fails current-approval support.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Context limits and evidence selection. Proposed explanation: “Token count equals word count”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The lesson uses supplied counts; actual counts depend on tokenizer and protocol.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Attention as weighted information selection

Target: Calculate a weighted combination of value vectors

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Calculate the output from the three supplied values and weights.

Teacher check: Teacher explanation: Contributions $[1,0]$, $[0,1]$, $[1,0.5]$ sum to $[2,1.5]$.

Individual check: For Attention as weighted information selection, use its worked case to evaluate this proposed explanation: “Attention copies the highest-weight token”. Explain the deciding step.

Expected correction: The supplied procedure combines vectors from multiple positions.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Can A alone justify answering 20 seats after the exception is known?

Reasoned teacher answer

No. The relevant exception limits this event to 12; omitting it would knowingly lose decisive evidence.

Changed-case reasoning: A+B costs 48 and no longer fits the 45-token evidence capacity. Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-evidence response.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Tokenize names, punctuation, emojis, and compound words; then hide different context cards and compare predicted continuations.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Token count equals word count: The lesson uses supplied counts; actual counts depend on tokenizer and protocol.

The smallest set is the best evidence: A set can fit while lacking current support for the requested claims.

Included means verified: Generation may misread or ignore evidence; verify claims after generation.

Check these misconceptions

Attention copies the highest-weight token: The supplied procedure combines vectors from multiple positions.

Deleting one contribution preserves a normalised mixture: Renormalise permitted mass under the stated contract.

Attention weights are truth probabilities: They are coefficients derived from compatibility, with no source-truth guarantee.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Nine exceeds eight. Shorten or select material while preserving essential instructions and evidence. Real systems also have output-budget rules.

Independent question

A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

Expected independent answer

The centimetres instruction may be outside the available context. Restate the required unit in the current request and check the output.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “All earlier conversation remains available forever.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Total=14, exceeding the toy limit by 2. Select a relevant 5-token source extract or shorten other material while retaining essential instructions and evidence. Real systems specify their own context and output limits; this toy policy must not be assumed universal.

Explanations for the additional workbook examples

W1 Compute available evidence

Under total 100, instructions 20, question 10 and output reserve 20, calculate evidence capacity.

Teacher explanation: $100 - 20 - 10 - 20 = 50$ supplied token units.

Separate the total budget into its declared parts: 100 minus 20 instruction units minus 10 question units minus 20 reserved output units leaves 50 units for evidence. Preserve the output reserve before filling evidence space. These are the lesson’s supplied token units, not a universal word-to-token conversion.

W2 Choose complete support

Compare A+B, A+C, C+B for fit and support of current approval and capacity.

Teacher explanation: A+B costs 50 and supports both. A+C costs 55 and fails budget; C+B costs 45 but uses old approval and fails current-approval support.

W3 Preserve a qualifier

A passage says room B normally holds 12, except during maintenance when the approved limit is 8. Why is retaining only the first clause inadequate?

Teacher explanation: The exception changes the applicable capacity during maintenance. Preserve conditions and determine whether maintenance applies; do not report 12 unconditionally.

W4 Budget new evidence independently

Total 120, instructions 25, question 15, output reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18. Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

Teacher explanation: Evidence capacity 50. D+E costs 50 and supplies both current claims. E+F costs 40 but old approval is insufficient. Without D, report the supported capacity and ask for current approval rather than infer it.

W5 Audit inclusion

A team says every claim is verified because all chosen documents fit in context. What further check is needed?

Teacher explanation: Check each generated claim against the actual source passage/version and preserve qualifiers. Budget fit and inclusion do not establish faithful use or current correctness.

Attention as weighted information selection — W1 Identify the roles

Explain query, key, value and weight without calling the largest weight a truth score.

Teacher explanation: The query compares keys, compatibility produces weights, and weights combine values. A weight is a computational coefficient, not a truth score.

Attention as weighted information selection — W2 Compute the mixture

Calculate the output from the three supplied values and weights.

Teacher explanation: Contributions $[1,0]$, $[0,1]$, $[1,0.5]$ sum to $[2,1.5]$.

Multiply each coordinate by its value's weight: $0.5*[2,0]=[1,0]$, $0.25*[0,4]=[0,1]$, and $0.25*[4,2]=[1,0.5]$. Add first coordinates to get $1+0+1=2$ and second coordinates to get $0+1+0.5=1.5$. The combined vector is $[2,1.5]$, not a literal selection of one source token.

Attention as weighted information selection — W3 Mask and renormalise

Exclude V3 and calculate permitted weights and output. Explain the empty-mask case.

Teacher explanation: Retained total 0.75 gives weights $2/3, 1/3, 0$ and output $[4/3, 4/3]$. All-excluded gives zero total and must report no available value under this contract.

Attention as weighted information selection — W4 Combine a new case independently

Values $A=[4,1]$, $B=[0,5]$, $C=[2,3]$; weights 0.25, 0.5, 0.25. Compute the full output. Mask B, renormalise the remaining weights and compute the new output.

Teacher explanation: Full contributions $[1,.25]$, $[0,2.5]$, $[.5,.75]$ give $[1.5,3.5]$. Retained total .5 gives A/C weights .5,.5 and output $[3,2]$. Masking changes available information; it does not verify sources.

Attention as weighted information selection — W5 Audit an explanation

A student claims the highest attention weight explains the entire model and authenticates a retrieved instruction. What additional reasoning is needed?

Teacher explanation: Inspect values and downstream computations, other heads and residual paths before explaining the whole model. Source truth and instruction authority require separate evidence and permissions; weight alone supplies neither.

Assess independent transfer

Selected case: Total 120, instructions 25, question 15, output reserve 30. Source D is current approval,

cost 28; E is current capacity, cost 22; F is old approval, cost 18. Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

Selected case answer

Teacher explanation: Evidence capacity 50. D+E costs 50 and supplies both current claims. E+F costs 40 but old approval is insufficient. Without D, report the supported capacity and ask for current approval rather than infer it.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

Keep the learner's question, approved source excerpt, and output rules inside the active context.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.
Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 4: Sampling and Variability. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it. Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires 50, exact D+E fit, E+F inadequacy and an explicit missing-approval fallback. Fresh retry: total 90, instructions 20, question 10, reserve 20 gives 40; current approval costs 24 and capacity 16, exactly fit. A cost-12 old approval does not replace current support.

Week 11 writes response specifications with explicit evidence requirements, output fields and fallback behaviour.

Readiness and reassessment for Attention as weighted information selection

Provide labelled intermediate columns and allow a calculator. Ask students to state the meaning and denominator of a value before interpreting it. Extension: implement the classroom calculation locally and test boundary cases without using personal data.

W4 requires [1.5,3.5] then [3,2] and explicit renormalisation. Fresh retry: A=[2,2], B=[6,0], C=[0,4], weights .5,.25,.25. Full output [2.5,2]; exclude A, remaining weights .5,.5 and output [3,2].

Plan the delayed return

Suggested return points in this level: ai-3/4, ai-3/6, ai-3/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its context budget.

7. Prepare, teach and assess

Teaching target: Account for fixed text, evidence and reserved output under a toy budget

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Compare A+B, A+C, C+B for fit and support of current approval and capacity.

Reasoned practice answer: Teacher explanation: A+B costs 50 and supports both. A+C costs 55 and fails budget; C+B costs 45 but uses old approval and fails current-approval support.

Check every learner before discussion: Use the worked case for Context limits and evidence selection. Proposed explanation: “Token count equals word count”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The lesson uses supplied counts; actual counts depend on tokenizer and protocol.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Increase B’s cost to 30 with all other costs fixed. Require a prediction and a reason before comparing. Expected result: A+B costs 48 and no longer fits the 45-token evidence capacity. Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-evidence response.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.